



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER



Test Report No. 2023 – 0041



Agricultural Machinery – Disc Harrow ACT 14 x 22

Test Report Number	2023 – 0041
Date of Issuance	February 22, 2023
Valid Until	February 22, 2028
Agricultural Machinery	Disc Harrow
Type	Trailing, Offset gang
Brand	ACT
Model	14 x 22
Test Requested by	ACT (MACHINERIES AND METAL CRAFT) CORPORATION Turayong, Cauayan City, Isabela

SUMMARY

Construction Criteria	Requirement as per PAES 120:2001	AMTEC Test
Length, mm	2000 to 8000	4170
Width, mm	1000 to 10000	1568
Height, mm	1000 to 1500	1200
Number of Gangs	Minimum of 2 (offset and single-action disc harrow)	2
Type of Disc	Plain or notched	Plain
Number of Discs Per Gang	Minimum of 4	7
Diameter of Discs, mm	400 to 650 (600 maximum for four-disc gang)	560
Thickness of Disc, mm	4 to 6	5.09
Concavity (Depth), mm	100 to 300	55
Disc Angle, °	24 maximum	Front (20.2°) Back (24.3°)
Length of Spool, mm	175 or 225 ± 2	220
Ground Clearance, mm	125	125
Weight Per Disc, kg	25 to 70	NM
Drawbar Power Requirement Per Disc, kW	1 to 2.5	1.62 ^a

^a Drawbar power requirement is computed as 86% of the maximum PTO power (26.44 kW) from the MASSEY FERGUSON MF1540 Four-Wheel Tractor performance test TR 2022-0697 divided by the number of discs.

Note: NM – Not Measured

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Not Provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly and installation	Provided
h. Specifications	Provided
i. Dismantling and disposal	Provided
j. Warranty	Not provided
k. Parts list	Not provided
2. Previous test reports of similar brand and model	2019 – 0214, 2019 – 0215, and 2023 – 0040
3. Manufacturer	ACT (MACHINERIES AND METAL CRAFT) CORPORATION
4. Four-wheel tractor used	29.9kW MASSEY FERGUSON MF1540
5. Serial number	142-2023
6. Number of operators	One
7. Handling and stability when machine is working	The disc harrow hitched to the tractor was stable during operation.
8. Handling and stability when machine is turning	The disc harrow hitched to the tractor was stable during turning
9. Quality of finished field: a. Tilth b. Level	The disc harrow produced good tilth (loose and friable). The level of the field was uniform.
10. Non-adhesion of soil to disc	Minimal adhesion of soil to the discs was observed.
11. Replacing and adjusting of parts	Some parts can be replaced and adjusted. No parts were replaced and adjusted during the test operation.
12. Durability of parts (based on wear of soil-working parts, visible deformation, etc)	There was no visible deformation on the components during and after the test operation.
13. Safety features	None
14. Failure or abnormalities that may be observed on the tractor or its component parts	No breakdown occurred during the test operation.

OBSERVATIONS (Cont'n)

15. Mild steel shall be used in the manufacture of the frame, gang axle, scraper, gang angling mechanism, and hitch.	Tubular steel (main frame) and channel bar (gang frame) and steel (gang shaft) are used in the manufacture of some of the components of the machine.
16. Cast iron shall be used in the manufacture of spool.	It was not verified if cast iron was used in the manufacture of the spool.
17. Carbon steel shall be used in the manufacture of hitch pin.	It was not verified if carbon steel was used in the manufacture of the spool.
18. Carbon steel with at least 80% carbon content (e.g. AISI 1080) or alloy steel with at least 0.0005% boron content shall be used in the manufacture of the disc blades.	It was not verified if carbon steel with at least 80 % carbon or alloy steel with at least 0.0005 % boron content was used in the manufacture of the disc blades.
19. The frame shall be rigid and durable.	The frame is rigid and stable.
20. The gangs shall be so attached that their angles can be easily changed to desired position with gang angling mechanism.	The gang angle can be changed to desired position with gang angling mechanism.
21. The rear gang(s) in offset and tandem disc harrows shall be so arranged that the disc(s) of rear gang(s) shall not make the same path formed by front gang(s).	Front and rear gangs are in offset and the disc of the two gangs were oriented opposite each other.
22. Bumpers may be provided in the inside end of the front gangs of the single-action and tandem disc harrows to prevent the inside discs from rubbing each other at any gang angle in which the gangs are set.	Not applicable
23. The discs in each gang shall be firmly fixed by spools and there shall not be any relative movement between the disc and the axle.	A spool is placed between the discs to prevent lateral movement of the shaft.
24. The scrapers shall be set in such a way that they shall not touch the face of disc and shall be able to scrape the soil effectively. Arrangement for adjusting the scrapers shall be provided.	The scrapers do not touch the face of the disc and effectively removed the soil clumps that adhered to the disc. No adjustment can be done on the scrapers since it was welded.
25. Adequate arrangement for lubrication of bearings shall be provided. The bearings shall be reasonably dust-proof and shall be properly aligned.	The bearings are provided with lubrication points. It was not verified if it is dust-proof and aligned.

OBSERVATIONS (Cont'n)

26. Ball or tapered roller bearings with special dirt seals shall be used	Ball bearings are used in the machine.
27. Both faces of spool coming in contact with the disc face shall be finished for proper gripping.	The spools in between discs are properly finished. They prevent lateral movement of the discs.
28. The hitch of disc harrow shall be compatible with the hydraulic system and the three-point hitch of the tractor specified in PAES 118.	The drawbar hitch of the disc harrow is compatible with that of the tractor
29. In single-action disc harrow, provision may be made for attaching a shovel or a sweep to cut and uncut the soil between the two gangs.	Not applicable
30. The harrow shall be easy to operate such as:	
a. Hitching to and unhitching from tractor;	Hitching and unhitching was easy.
b. Adjusting the depth of cut;	Depth of cut can be adjusted through the depth adjustment and hydraulic-operated pneumatic wheels.
c. Changing the position of the harrow with respect to the line of pull of the tractor;	Not verified
d. Moving the standards laterally on the frame;	Not verified
e. Varying the angle between the bodies;	The gang angle can be adjusted through the gang angling mechanism.
f. Maneuverability during operation;	The tractor with the hitched disc harrow was stable during turning.
g. Clearing blockages;	No blockages were observed during test operation.
h. Changing from transport to work position and vice versa; and	Changing the position from transport to work and vice versa can be done through the four-wheel tractor's hydraulic cylinder which raised or lowers the pneumatic wheels.
i. Adjustment of the scrapers.	Scrapers are welded and cannot be adjusted.
31. Others remarks	There were no warning notices or markings. Only the rear reflective stickers and nameplate of disc harrow were provided.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing / Commercial
Date of Test	January 31, 2023
Location of Test	Brgy. Villa Miguel, Quirino, Isabela 17°8'14.31" N, 121°43'8.63" E
Items Tested	Operator's Manual, Width of Harrowing, Field Capacity, Field Efficiency, Noise Level, Number of Operators, and Fuel Consumption Rate

METHODS OF TEST

The disc harrow was tested following the PAES 133:2004 – Agricultural Machinery – Disc Harrow – Methods of Test.

DESCRIPTION OF THE MACHINE



Figure 1. Parts of ACT 14 x 22 Disc Harrow hitched to 29.9kW MASSEY FERGUSON MF1540 include the following: (a) Four-wheel tractor, (b) Hydraulic cylinder, (c) Main frame, (d) Transport wheel, (e) Hitch point, (f) Disc, (g) Spool, and (h) Scraper

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Overall dimensions and weight		
1.1	Length, mm	4100	4170
1.2	Width, mm	1630	1568
1.3	Height, mm	930	1200
1.4	Weight, kg	690	NM
1.5	Weight per disc, kg	8.8	NM
2	Ground clearance, mm	ND	125
3	Gang		
3.1	Quantity	2	2
3.2	Number of discs per gang	9	7
3.3	Disc spacing, mm	220	220
3.4	Angle of front gang, °	14.98	20.2
3.5	Angle of rear gang, °	17.74	24.3
3.6	Size and shape of gang shaft, mm	38.1 (Solid circular shaft)	NM (Solid circular shaft)
4	Disc		
4.1	Brand	NIAUX	BULLS EYE
4.2	Make	France	ND
4.3	Type	Plain	Plain
4.4	Diameter, mm	559	560
4.5	Thickness, mm	4	5.09
4.6	Concavity, mm	75	55

^a The data were obtained from the specifications sheet and operator's manual provided by the requesting party.

Note: ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
5	Scraper		
5.1	Length, mm	100	140
5.2	Width, mm	85	83
5.3	Thickness, mm	7.00	9.56
6	Spool		
6.1	Length, mm	220	220
6.2	Diameter, mm	74.67	75.81
6.3	Shape and size of hole, mm	Spool, ND	Circular, NM
6.4	Material	Cast iron	ND
7	Main frame		
7.1	Dimension, L × W, mm	2850 × 835	2750 × 840
7.2	Material (L × W × t, mm)	Square tube (76.2 × 76.2 × 6.35)	Square tube (80 × 80 × NM)
8	Transport wheels		
8.1	Quantity	2	2
8.2	Dimensions, D × W, mm	6.00-14LT	690 × 190 (6.00-15LT)
9	Bearings		
9.1	Type	Ball bearing	Ball bearing
9.2	Brand	Koyo	ND
9.3	Quantity per gang	2	2
10	Additional weight, kg	ND	ND
11	Operating width, mm	1400	1400

^a The data were obtained from the specifications sheet and operator's manual provided by the requesting party.

Note: ND – No Data, NM – Not Measured

RESULTS

Item		Data	
1	Test conditions		
1.1	Type of field operation	Secondary Tillage – Harrowing	
1.2	Condition of field		
1.2.1	Location	Quirino, Isabela	
1.2.2	Dimensions of field, L × W, m	44.0 × 24.0	
1.2.3	Area, m ²	1056.0	
1.2.4	Soil type	Silty clay loam	
1.2.5	Soil moisture content, % _{db}	25.40	
1.2.6	Height of stubbles, mm	NA	
1.2.7	Spacing of stubbles, row × hill, mm	NA	
1.2.8	Weed density	Low	
1.2.9	Soil hardness, kg/cm ²	NM	
1.2.10	Last crop planted	Corn	
1.2.11	Field condition before operation	Plowed	
2	Field performance	1 st pass	2 nd pass
2.1	Brand and model of tractor used	MASSEY FERGUSON 1540	
2.2	Tractor's gear shift setting	High 1 st Gear	
2.3	Traveling or operating speed, kph	8.54	8.02
2.4	Average depth of harrowing, mm	NM	NM
2.5	Average width of harrowing, mm	1361	1314
2.6	Operating width, mm	1400	1400
2.7	Actual field capacity		
2.7.1	ha/h	1.14	1.07
2.7.2	h/ha	0.881	0.938
2.8	Theoretical field capacity		
2.8.1	ha/h	1.20	1.12
2.8.2	h/ha	0.836	0.890
2.9	Field efficiency, %	94.90	94.80
2.10	Fuel consumption rate, L/h	7.13	8.42
2.11	Specific fuel consumption, L/ha	6.17	7.93
2.12	Field operational pattern	Headland alternating pattern	Headland alternating pattern
2.13	Overlapped, %	12.50	10.91
2.14	Noise level, dB(A)	90.4	90.9

Note: NA – Not Applicable, NM – Not Measured

OBSERVATIONS



Figure 2. Quality of Work.



Figure 3. Nameplate of the ACT 14 × 22 Disc Harrow.

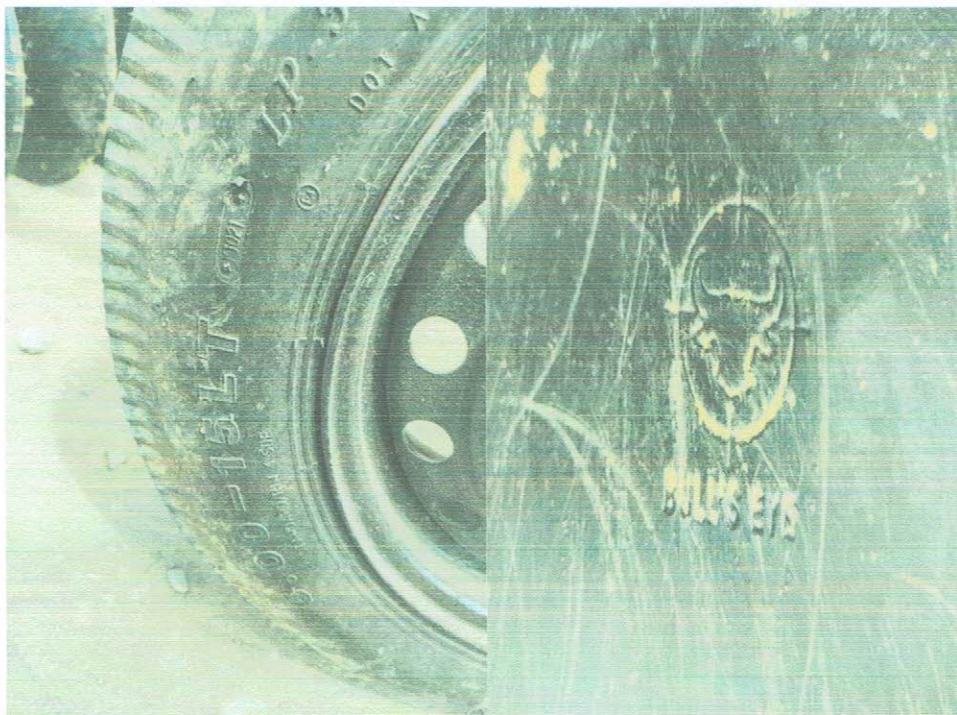
OBSERVATIONS (Cont'n)

Figure 4. Tire size (left) and Markings on the disc (Right) of ACT 14 x 22 Disc Harrow.

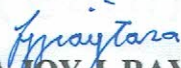


Figure 5. Reflective stickers on the ACT 14 x 22 Disc Harrow.

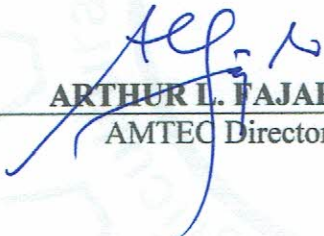
Tested by:


DEO-JAY T. MANGLAL-LAN
Test Engineer

Verified by:


FATIMA JOY J. RAYTANA
Test Engineer

Approved for Release:


ARTHUR L. FAJARDO
AMTEC Director

MAR 07 2023
Date

REMINDER

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